Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (W), Mumbai : 400058, India

(An Empowered Autonomous College Affiliated to University of Mumbai)

Special Exam Examination

July 2025

Max. Marks: 100 | Duration: 3 hours

Class: Third Year | Semester: V

Course Code: CS303 | Program: CE

Name of the Course: Artificial Intelligence and Machine Learning

Instructions:

(1) All questions are compulsory

(2) Draw neat diagrams and write equations wherever necessary

(3) Assume suitable data if necessary

Course Outcomes (CO): At the End of the course students will be able to

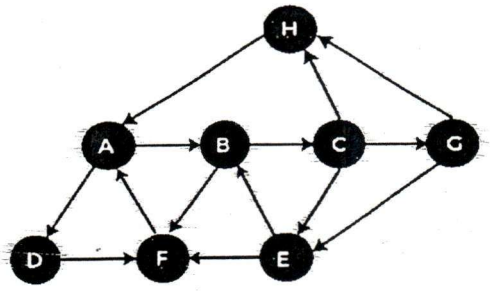
CO1 Understand AI building blocks presented in intelligent agents

CO2 Solve the problems using suitable searching methods.

CO3 Solve the problems using suitable reasoning and knowledge representation methods.

CO4 Apply suitable machine learning technique for a given problem

CO5 Design an intelligent system using different AIML techniques for real life problems.

Q. No		Max. Marks	CO-BL
Q.1a	Give a PEAS description of the task environment and characterize it in terms of the properties for Automated Car Drive, Part-picking robot, Chess Playing-AI, Spam Filter.	10	1-3
Q.1b	State the relationship between agents and environment. Explain the concept of Rationality. Explain types of environments.	10	1-2
Q.2a	Use Depth First Search algorithm to the graph given below lexicographically and write the path from A to reach the goal node G. Write about the time and space complexity of DFS. 	10	2-3

<table>
<tr><td>Q.2b</td><td>Write an algorithm for steepest hill climbing. Give the scenario where it will work and where it will fail.</td><td>10</td><td>2-3</td></tr>
<tr><td>Q.3a</td><td>Represent the following statements in predicate logic:
i) Marcus tried to assassinate Caesar.
ii) All Pompeian's were Roman.
iii) All Romans were either loyal to Caesar or hated him.
iv) Everyone is loyal to someone.
v) People only try to assassinate rulers they are not loyal to.</td><td>10</td><td>3-3</td></tr>
<tr><td>Q.3b</td><td>consider the following sentences
• John likes all kinds of food
• Apples are food
• Chicken is food
• Anything anyone eats and isn't killed by is food
• Bill eats peanuts and is still alive
• Sue eats everything bill eats
i) Translate these sentences into formulas in predicate logic
ii) Prove that john likes peanuts using backward chaining
iii) Convert into clause form
iv) Prove that john likes peanuts using resolution</td><td>10</td><td>3-4</td></tr>
<tr><td>Q.4a</td><td>What is supervised learning? How can it be used for anomaly detection and market segmentation?
OR
Consider the following data set consisting of the scores of two variables on each of seven individuals:
<table><tr><th>Subject</th><th>A</th><th>B</th></tr><tr><td>1</td><td>1.0</td><td>1.0</td></tr><tr><td>2</td><td>1.5</td><td>2.0</td></tr><tr><td>3</td><td>3.0</td><td>4.0</td></tr><tr><td>4</td><td>5.0</td><td>7.0</td></tr><tr><td>5</td><td>3.5</td><td>5.0</td></tr><tr><td>6</td><td>4.5</td><td>5.0</td></tr><tr><td>7</td><td>3.5</td><td>4.5</td></tr></table>This data set is to be grouped into two clusters. Point 1 and 4 are the seed points. Use the Euclidean distance measure.</td><td>10</td><td>4-3</td></tr>
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Q.4b	How does the KNN algorithm work? What is the significance of the 'k' parameter in KNN? Can KNN be used for regression tasks?	10	4-4
Q.5a	How to Handle Missing Data in Linear Regression? How do you determine if your linear regression model is overfitting?	10	4-4
Q.5b	What is regularization, and how does it relate to the bias-variance tradeoff? How can you reduce bias in a model?	10	4-4